ANNEXURE 4.1

SMART SPECTACLES FOR VISUALLY IMPAIRED PEOPLE

A project report submitted in partial fulfilment of the requirement for the Award of the Degree of

BACHELOR OF ENGINEERING

in

Electronics and Communication Engineering

By

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Under the Guidance of

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Professor

Estd : 2008

Department of Electronics and Communication Engineering

Methodist College of Engineering and Technology (Autonomous)

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2025-26

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DECLARATION BY THE CANDIDATES

We, **N V S Pradyumna, 160722735078, B Nikhil Sai, 160722735084** and **B Mohith, 160722735120** students of Methodist College of Engineering and Technology, pursuing Bachelor's degree in Electronics & Communication Engineering , hereby declare that this Project report entitled "**Smart Spectacles for Visually impaired people",** carried out in 2025-26 under the guidance of **Dr K Jaya Sankar, Professor.** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Electronics & Communication Engineering. This is a record work carried out by us and the results embodied in this report have not been reproduced/copied from any source.

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CERTIFICATE BY THE SUPERVISOR

This is to certify that this Dissertation on project work entitled “**Smart Spectacles for Emergency Rescue” By N V S Pradyumna, 160722735078, B Nikhil Sai, 160722735084 And B Mohith, 160722735120** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Electronics & Communication Engineering, during the academic year 2025-26, is a bonafide record of work carried out by them.

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Dr. VSSN Srinivasa Baba,
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PROJECT APPROVAL CERTIFICATE

This is to certify that this Dissertation on project work entitled "**Smart Spectacles for visually impaired people" By N V S Pradyumna, 160722735078, B Nikhil Sai, 160722735084 and B.Mohith, 160722735120** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Electronics & Communication Engineering of the Osmania University, Hyderabad, during the academic year 2025-26, is a bonafide record of work carried out by them.

Internal Examiner **External Examiner** **HOD,ECE**

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PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
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PO8: Individual and collaborative teamwork: Function effectively as an individual, and as a member or leader in diverse teams/multi-disciplinary teams.
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PO10: Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11: Life-long learning: Recognize the need for and have the preparation and ability for i). independent and life-long learning ii). Adaptability to new and emerging technologies and iii). Critical thinking in the broadest context of technological change

Abstract

Smart Spectacles for Emergency Rescue is an innovative wearable assistive technology developed to enhance the mobility, safety, and situational awareness of visually impaired and elderly individuals. Conventional mobility tools such as white canes offer limited functionality, as they primarily detect ground-level obstacles and cannot identify head-level hazards, moving objects, or emergency situations. Commercial smart glasses that attempt to provide enhanced perception are often expensive, bulky, and lack integrated features such as fall detection, autonomous alert systems, and real-time location tracking. To address these limitations, this project proposes a compact, low-cost, and intelligent spectacle-based system that integrates advanced sensing, artificial intelligence, and Internet of Things (IoT)-enabled communication into a single wearable platform.

The proposed system utilizes an ESP32-CAM/ESP32-S3 microcontroller for processing and embedded vision, a VL53L1X Time-of-Flight (ToF) sensor for precise obstacle detection, an ADXL345 inertial measurement unit (IMU) for motion monitoring and fall detection, a TELEGRAM API KEY for autonomous emergency message transmission, and a u-blox M9 GPS module for accurate geolocation. The ToF sensor continuously measures the distance of surrounding obstacles and provides real-time alerts to the user through a haptic feedback mechanism when an object is detected within a predefined threshold. The IMU intelligently detects abnormal motion patterns and impact signatures that indicate a fall or medical emergency. In such scenarios, the system automatically retrieves the user's GPS coordinates and transmits an emergency SMS alert to pre-configured caregivers, thereby ensuring timely assistance even when the user is unconscious or unable to seek help.

Additionally optional AI-based object recognition through the ESP32-CAM module, enabling detection of common hazards such as staircases, vehicles, and walls to further improve user awareness. Experimental testing demonstrated that the device successfully detected obstacles within a 45–50 cm range with high reliability, delivered immediate haptic feedback, accurately recognized common objects, dispatched emergency SMS alerts within 5–6 seconds following fall detection, and maintained stable GPS connectivity. Battery performance tests confirmed an operating duration of approximately 8–10 hours under continuous use, making the device suitable for daily practical deployment.

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1. Introduction

In this chapter, an overview of the Smart Spectacles for Visually Impaired People was presented, highlighting the challenges faced by visually impaired and elderly individuals in navigating complex environments safely and independently. The limitations of traditional mobility aids such as white canes and guide dogs were discussed, emphasizing their inability to detect elevated obstacles or communicate emergencies. The proposed system integrating a VL53L0X Time-of-Flight sensor, ADXL345 accelerometer, NEO-6M GPS module, and ESP32 WiFi-based Telegram Bot API was introduced as a comprehensive wearable solution. The need for a low-cost, compact, and practically deployable assistive device suitable for daily use was also established, along with the system's modularity for future enhancements such as voice assistance and AI capabilities.

1.1 Overview for Smart Spectacles for Visually Impaired People

The Smart Spectacles for Emergency Rescue project presents an integrated wearable assistive solution designed to enhance the mobility, safety, and independence of visually

impaired and elderly individuals. Traditional mobility aids such as white canes provide basic tactile feedback but fail to detect elevated obstacles, identify real-world hazards, or initiate emergency communication during critical events. This project addresses these limitations by incorporating advanced sensing technologies, AI-enabled perception, and automated communication mechanisms into a compact, lightweight spectacle frame.

The system operates through multiple coordinated modules. A VL53L1X Time-of-Flight (ToF) sensor continuously measures the distance to nearby obstacles with millimetre accuracy, offering reliable detection even under varying lighting conditions. An LADXL345 Sensor monitors user movement and identifies abnormal motion patterns, such as sudden impacts or falls. In the event of a fall, the system automatically acquires the user’s geographical coordinates via the GPS module and sends an emergency alert using the ESP32 built-in WiFi with Telegram Bot API to predefined caregivers.

1.1.1 Need for Smart Spectacles for Visually Impaired People

The need for Smart Spectacles for Emergency Rescue arises from the growing challenges faced by visually impaired and elderly individuals in navigating their daily environments safely and independently. Traditional mobility aids such as white canes, walking sticks, or guide dogs offer only partial situational awareness and fail to address the broader spectrum of hazards encountered in modern environments. These limitations highlight the necessity for a more advanced, technology-driven assistive device capable of providing real-time environmental perception and automatic emergency support.

Another significant need arises from the heightened risk of **falls**, especially among elderly or mobility-impaired individuals. Falls can lead to severe injuries, unconsciousness, or medical emergencies where the user may be unable to call for help. A built-in IMU enables the detection of falls or abrupt impacts and automatically triggers an alert mechanism. This feature ensures that timely assistance can be provided, significantly reducing response times during emergencies

Finally, existing smart wearable technologies designed for assistive navigation are often **expensive, bulky, or inaccessible**, limiting their adoption among populations in developing regions. Smart Spectacles for Emergency Rescue address this need by using **low-cost, energy- efficient components** such as the ESP32-S3 or ESP32-CAM, making the solution affordable without compromising functionality.

1.1.2 Limitations of Existing Systems

Assistive technologies used by visually impaired and elderly individuals—such as white canes, guide dogs, ultrasonic navigation devices, and commercial smart wearables—provide partial support but fall short of delivering comprehensive safety and real-time emergency assistance. These limitations create a significant gap in user independence, environmental awareness, and rapid emergency response. The major limitations of current systems are outlined below:

1. **Traditional White Cane**

The white cane is the most widely used mobility aid; however, it has several major shortcomings:

- Detects only ground-level obstacles and does not identify elevated hazards such as signboards, tree branches, or vehicle mirrors.
- Cannot detect fast-approaching objects, especially moving vehicles or dynamic obstacles.
- Provides no information about the shape, size, or type of an obstacle.
- Offers no emergency alert mechanism, leaving users vulnerable during falls or medical incidents.

2. **Ultrasonic-Based Assistive Devices**

Many existing electronic travel aids use ultrasonic sensors for obstacle detection, but they introduce several technical issues:

- Poor accuracy at long distances or with small, thin obstacles.
- Difficulty detecting transparent or angled surfaces, leading to false negatives.
- Ultrasonic signals are prone to environmental noise interference, reducing reliability.
- Limited range (typically 1–2 meters), which may not provide sufficient reaction time.
- Limited or no integration with emergency alert systems.

3. Smartphone-Based Navigation Apps

GPS-based mobile apps offer route guidance but are not suitable for visually impaired users in all situations:

- Reliance on internet connectivity, which may not always be available.
- Cannot detect obstacles or identify hazards in real time.
- Require user interaction, which is difficult during emergencies or when hands-free operation is necessary.
- No fall detection or automated SMS alerting.

4. Commercial Smart Glasses

Some advanced smart glasses are available, but they come with major limitations:

- Very high cost, making them unaffordable for the majority of visually impaired individuals.
- Often bulky or uncomfortable, reducing long-term usability.
- Limited in features such as fall detection, AI-based recognition, or GSM-based emergency communication.
- Dependency on external devices or cloud services, increasing complexity and reducing reliability

1.2 Evolution of Smart Spectacles for Emergency Rescue

The evolution of Smart Spectacles for Emergency Rescue is rooted in the continuous advancement of assistive technologies designed to support visually impaired and elderly individuals. Early assistive devices primarily consisted of simple mechanical tools such as white canes, which offered only tactile ground-level obstacle detection and lacked the ability to sense elevated hazards or dynamic environmental changes. As technology progressed, electronic travel aids emerged using ultrasonic sensors to detect obstacles, but these systems often suffered from limited accuracy, short detection ranges, and susceptibility to noise interference, making them unreliable in complex outdoor environments. The introduction of embedded microcontrollers marked a significant step forward, enabling wearable devices to process sensor data and deliver real-time feedbackThe next major breakthrough came with lightweight artificial intelligence and edge computing platforms like ESP32 and ESP32-S3, which made it possible to perform on-device object recognition and decision-making without requiring cloud resources. Simultaneously, high-precision sensors such as the VL53L1X Time- of-Flight sensor and ADXL345 enabled accurate obstacle detection and fall monitoring in compact wearables. These technological advancements collectively shaped the modern concept of Smart Spectacles—an integrated system that combines obstacle detection, fall detection, GPS tracking, and automated emergency alerts within a compact, low-cost, and user-friendly eyewear design. Furthermore, the miniaturization of electronic components and improvements in battery efficiency have made it possible to design lightweight, comfortable eyewear that can be worn throughout the day without discomfort. The integration of multi-sensor data fusion techniques has significantly improved the accuracy and reliability of environmental perception in dynamic and crowded spaces. As safety requirements increased for vulnerable populations

1.3 Role of IoT in Smart Spectacles

The Internet of Things (IoT) plays a central and transformative role in the functionality of Smart Spectacles for Emergency Rescue by enabling seamless integration, communication, and intelligent decision-making across all system components. IoT allows the various sensors—such as the Time-of-Flight (ToF) sensor, Inertial Measurement Unit (IMU), GPS module, and ESP32 processor—to operate as interconnected nodes that continuously collect, process, and exchange data in real time. Through IoT-based architecture, the ESP32 microcontroller acts as the central hub, interpreting sensor outputs, detecting emergencies, and coordinating responses without requiring user intervention. The IoT-enabled SIM7600 GSM/LTE module is responsible for transmitting emergency alerts, including the user's live GPS coordinates, to caregivers or assigned contacts, ensuring immediate communication even in the absence of internet connectivity. It also supports scalability, allowing additional sensors or features to be incorporated without major redesign. By enabling continuous connectivity, real-time responsiveness, automated alerting, and intelligent environmental awareness, IoT significantly enhances the performance, reliability

1.4 Motivation

The motivation behind developing Smart Spectacles for Emergency Rescue stems from the urgent need to provide visually impaired and elderly individuals with a safer, more independent, and technologically empowered way to navigate their surroundings.

Traditional mobility aids such as white canes offer limited assistance as they only detect obstacles at ground level and fail to alert users about elevated hazards or rapidly approaching dangers. This lack of comprehensive situational awareness often places users at heightened risk of collisions, injuries, and reduced confidence in mobility. Moreover, elderly individuals and people with health vulnerabilities face an increased likelihood of falls, which remain one of the leading causes of serious injuries and medical emergencies.

In many fall-related incidents, timely help is delayed because the individual is unable to call for assistance, leading to further complications. Existing assistive devices do not provide automated emergency communication, real-time location sharing, or intelligent hazard detection, leaving a critical gap in user safety.

With the availability of advanced sensors, IoT communication technologies, there is immense potential to develop a compact wearable solution capable of addressing these challenges. The motivation is further strengthened by the need to create an affordable, accessible, and easy-to-use device, especially for populations in developing countries

Smart Spectacles aim to bridge the gap between technological innovation and social responsibility by providing a reliable, life-saving device that improves the quality of life for vulnerable communities. Ultimately, this project is motivated by a commitment to create a meaningful, impactful, and human-centric

technological solution that empowers users to live with greater freedom, confidence, and safety.

1.5 Scope Of The Project

The scope of the Smart Spectacles for Emergency Rescue project encompasses the design, development, integration, and evaluation of a wearable assistive device that enhances the safety and independence of visually impaired and elderly individuals. This project aims to build a compact spectacle-mounted system capable of detecting obstacles, identifying fall events, transmitting emergency alerts, and providing real-time location tracking. The scope includes the incorporation of a VL53L1X Time-of-Flight sensor for precise obstacle detection, an ADXL345 Sensor for monitoring motion and fall patterns, a GPS module for geolocation, and a ESP32 built- in WiFi with Telegram Bot API for communication. The project further involves implementing haptic or auditory feedback mechanisms to alert users immediately when a potential hazard is detected.

The scope also covers the development of the software algorithms needed to process sensor data, evaluate risk conditions, and initiate emergency communication. This includes calibration routines for fall detection, threshold tuning for obstacle sensing, integration of real-time GPS positioning, and sending automated alert messages to designated contacts. The project aims to ensure low power consumption, portability, and continuous operation through appropriate hardware selection and power management strategies.

Evaluation of the project includes testing the system in various indoor and outdoor environments to verify its accuracy, responsiveness, and reliability. The testing phase also assesses user comfort, battery performance, detection accuracy, communication speed, and overall usability.

1.6 Applications

Navigation for visually impaired individuals:

Provides real-time obstacle detection, AI-based object recognition, and haptic feedback to support safe and independent mobility.

Safety support for elderly individuals:

Detects falls and automatically sends emergency alerts with GPS location to caregivers or family

Use in assisted-living and healthcare facilities:

Enables continuous monitoring of vulnerable individuals, ensuring timely response during medical or mobility-related emergencies.

Outdoor and unfamiliar area navigation:

Helps users navigate safely in crowded streets, public spaces, and environments where traditional aids are insufficient.

Search and rescue operations:

GPS tracking and alerting capabilities allow immediate location identification in emergency situations.

Support for individuals with cognitive impairments:

Assists users prone to disorientation or confusion by providing safety alerts and automated communication support.

Industrial safety applications:

Can be adapted to warn workers about hazards in low-visibility or high-risk environments, enhancing workplace safety.

IoT-based monitoring systems:

Acts as a wearable IoT node for collecting environmental and movement data for analytics and safety insights.

2. Literature Review

In this chapter, various existing research works and assistive technologies related to mobility and safety for visually impaired and elderly individuals were discussed. The limitations of traditional aids and electronic systems such as white canes, ultrasonic devices, and existing smart wearables were analyzed. Different approaches including laser-based detection, IMU-based fall detection, and IoT-based emergency systems were reviewed. A gap analysis was performed to identify the shortcomings of current solutions. Based on this, the objectives and problem statement for the proposed system were clearly defined.

2.1 Literature Survey

The development of assistive technologies for visually impaired and elderly individuals has been an active area of research for several decades, with numerous studies focusing on enhancing mobility, environmental awareness, and safety through the use of sensors, computer vision, IoT, and wearable devices. Traditionally, mobility assistance has relied on non-electronic tools such as white canes, which provide only tactile feedback and fail to detect elevated or fast-moving obstacles. Studies such as those by Benjamin et al. highlighted the limitations of white canes, emphasizing their inability to provide full 3D environmental awareness. This limitation led to the exploration of ultrasonic-based navigation systems, which became a foundation for early electronic travel aids (ETAs). Research by Kay and Hoyle demonstrated the effectiveness of ultrasonic sensors for short- range detection; however, later studies revealed challenges such as limited range, sensitivity to environmental noise, and difficulty detecting thin or transparent objects, prompting the need for more robust sensing technologies.

With advancements in microcontrollers and embedded systems, researchers began integrating infrared sensors, RFID systems, and inertial measurement units (IMUs) into wearable navigation aids. IMU-based fall detection systems gained prominence due to their ability to identify sudden acceleration changes, with studies by Bourke et al. demonstrating high accuracy using threshold- based methods. However, most IMU-based systems lacked automated communication capability, making them insufficient for emergency rescue scenarios. The introduction of Time-of-Flight (ToF) sensors marked a significant improvement in obstacle detection accuracy. Studies by Amaro et al. showed that ToF sensors such as VL53L1X outperform ultrasonic sensors in precision and responsiveness, especially in variable lighting conditions, making them suitable for wearable applications.

2.1.1 "A Laser Cane for the Blind"

- Introduced an early electronic mobility aid using laser-based obstacle detection.

- Demonstrated improved range and accuracy compared to traditional white canes.
- Focused on detecting obstacles at different heights, improving user safety.
- Lacked advanced features such as AI-based classification and emergency alerts.
- Highlighted the need for more compact, wearable, and multi-functional assistive technologies.

2.1.2 "Evaluation of a Threshold-Based Fall Detection Algorithm Using a Tri-axial Accelerometer"

- Developed a fall detection system using an IMU-based threshold algorithm.
- Demonstrated high accuracy for identifying sudden falls in elderly individuals.
- Highlighted the challenges of sensor calibration and false positives in daily activities.
- Did not include automated emergency communication or GPS tracking.
- Emphasized the need for integrated systems combining fall detection with rescue support.

2.1.3 "IoT-based Emergency Alert System for Elderly People"

- The study focuses on creating an IoT-based emergency alert system for elderly individuals using sensors and GSM modules.
- The system improves emergency response time by eliminating the need for manual intervention.
- It demonstrates the effectiveness of GSM in remote alerting where internet connectivity is not reliable.
- The study lacks obstacle detection, AI vision, wearable integration, and GPS tracking—gaps your project addresses

2.2 Gap Analysis

1. **From Laser Cane (Benjamin et al.)**

- Detects obstacles but is **not wearable**.
- **No fall detection** or emergency alert system.
- Cannot detect **moving objects** or provide GPS location.

Gap Filled: Smart Spectacles provide wearable ToF sensing, fall detection, and GPS- based emergency alerts.

2. **From IMU-Based Fall Detection (Bourke & Lyons)**

- Accurately detects falls using accelerometers.
- **No automatic SMS alerts** to caregivers.
- **No obstacle detection** or object recognition.

Gap Filled: Smart Spectacles add SMS alerts, GPS tracking, and obstacle detection features.

3. **From IoT Emergency Alert System (Sharma et al.)**

- Sends emergency SMS via GSM effectively.
- Does **not assist in mobility** (no obstacle detection).
- Not designed as a **wearable device**.

Gap Filled: Smart Spectacles integrate mobility support + GSM alerts + wearable design.

4. **From Ultrasonic Navigation Aids**

- Offer low-cost obstacle detection.
- Poor accuracy for **thin/transparent objects**.
- No **GPS or emergency alert** functionality.

Gap Filled: Smart Spectacles use ToF sensors for accurate detection and include emergency rescue features.

Overall Gaps Identified

- No single device provides **obstacle detection + fall detection + GPS + GSM alerts** together.
- Existing solutions are either bulky, expensive, or incomplete.
- Limited wearable options for visually impaired users.
- Lack of **AI-based hazard recognition** in low-cost devices.

What Your Project Solves

- Provides a **complete, wearable, low-cost** rescue solution.
- Integrates **ToF sensing, IMU fall detection, GPS, GSM alerts, and AI vision** in one device.

2.3 Objectives

The primary objective of the Smart Spectacles for Emergency Rescue project is to design and develop an intelligent, IoT-enabled wearable assistive device that enhances the safety, mobility, and independence of visually impaired and elderly individuals. The detailed objectives of the project are as follows:

- To design and implement a fall detection mechanism using an ADXL345 Sensor that analyzes acceleration and motion patterns to accurately identify sudden falls or abnormal movements.
- To integrate GPS functionality for real-time location tracking, especially during emergencies, enabling caregivers or responders to locate the user instantly.
- To implement ESP32 built-in WiFi with Telegram Bot API emergency communication through the SIM7600 module, allowing the system to automatically send SMS alerts with the user's live location upon detecting a fall or critical situation.
- To ensure seamless IoT connectivity and system coordination, enabling smooth interaction among sensors, microcontrollers, and communication modules for reliable real-time operation.
- To design a compact, lightweight, and low-power wearable form factor suitable for continuous daily use, ensuring user comfort and long battery life.
- To evaluate system performance through extensive indoor and outdoor testing to measure accuracy, reliability, response time, usability, and overall effectiveness in real-world conditions.

Overall, these objectives aim to deliver a fully functional, affordable, and user-friendly smart spectacle system that significantly improves safety, promotes independent navigation, and provides timely emergency assistance to vulnerable individuals.

2.4 Problem Statement

Visually impaired and elderly individuals face significant challenges in navigating their surroundings safely due to limited environmental awareness and the inability to detect elevated hazards, moving obstacles, or sudden changes in terrain. Traditional mobility aids such as white canes and ultrasonic devices provide only partial assistance and fail to identify many real-world risks. Additionally, elderly individuals are highly vulnerable to falls and medical emergencies, and in many cases, they are unable to call for help immediately, resulting in delayed assistance xisting assistive technologies are often expensive, limited in functionality, or dependent on external devices and stable internet connectivity, making them impractical for daily use in diverse environments. Despite numerous advancements in sensing and communication technologies, there is currently no affordable, compact, and system that combines obstacle detection, fall detection, GPS tracking, and automated emergency communication into a single device.

3. Design Analysis

In This Chapter,The Smart Spectacles for Emergency Rescue are designed as a compact wearable system that integrates sensing, processing, communication, and alert mechanisms into a single unit. The design focuses on providing real-time safety support to visually impaired and elderly individuals through accurate obstacle detection, reliable fall monitoring, and automated emergency communication.

The system uses a Time-of-Flight (ToF) sensor to detect obstacles in front of the user and provide immediate haptic or audio feedback. A 6-axis IMU sensor monitors body movement and identifies sudden falls using threshold-based analysis. The ESP32 microcontroller serves as the central processing unit, collecting data from sensors, making decisions, and controlling the communication modules.

3.1 Model Architecture

1 **Central Processing Unit – ESP32**

- Acts as the main controller of the system.
- Processes sensor data, executes logic, and controls communication modules.

2 **Sensing Layer**

- ToF Sensor (VL53L1X): Detects obstacles and measures distance in real time.
- IMU Sensor (ADXL345): Monitors acceleration and orientation to detect falls.

3 **Communication Layer**

- GPS Module: Provides real-time location during emergencies.
- ESP32 built-in WiFi with Telegram Bot API: Sends automated SMS alerts to caregivers with live location.

4 **Feedback Layer**

- Buzzer: Alerts the user immediately when an obstacle or hazard is detected.

5. Power Management

- Rechargeable battery powers all components.
- Voltage regulators ensure stable operation and low power consumption.

3.1.1 Proposed System Architecture

1 . Input Sensing Components

- Time-of-Flight (ToF) Sensor: Continuously measures distance to obstacles in front of the user.
- IMU Sensor (Accelerometer + Gyroscope): Detects sudden falls or abnormal motion.

2 . Core Processing Unit

- ESP32 Microcontroller:

o Reads sensor data and processes it in real time.

o Executes fall detection logic and obstacle detection algorithms.

o Controls communication modules and feedback outputs.

o Optional lightweight AI algorithms for object identification.

3 . Communication Modules

- GPS Receiver: Provides accurate latitude and longitude when an emergency occurs.
- ESP32 built-in WiFi with Telegram Bot API: Sends automated SMS alerts with the user's live location to caregivers or emergency contacts.

4. User Feedback Mechanisms

- Buzzer: Provides audio alerts during fall detection or hazardous conditions.

• Voltage regulation ensures stable 3.3V/5V supply to sensors and modules.

• Designed for low power consumption suitable for wearable use.

5. Integration Layer (Wearable Frame)

• All sensors, processors, and communication components are compactly mounted on spectacle frames.

• Ensures lightweight, ergonomic, and user-friendly operation.

6. Emergency Response Layer

• When a fall is detected, the system automatically:

  o Fetches GPS coordinates

  o Sends SMS alerts through ESP32 built-in WiFi with Telegram Bot API

• Ensures quick response even if the user is unconscious.

System Block Diagram :

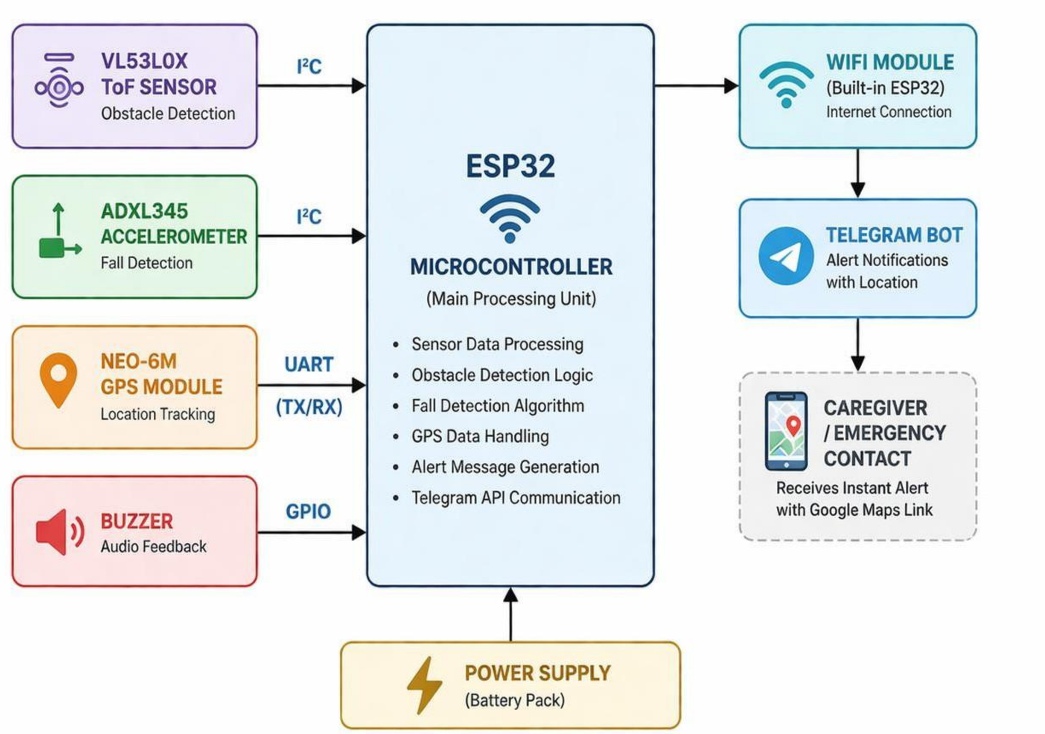

Fig. 3.1: System Block Diagram of Smart Spectacles for Visually Impaired People

3.1.2 System Components Description

1. **ESP32 Dev kit**

- Serves as the central processing unit of the system.
- Handles sensor data acquisition, decision-making, and communication control.

2. **Time-of-Flight (ToF) Sensor – VL53L1X**

- Measures distance using laser-based ranging technology.
- Provides high accuracy for obstacle detection up to several meters.
- Sends continuous proximity data to ESP32 for real-time hazard alerts.

3. **IMU Sensor – ADXL345**

- Combines accelerometer and gyroscope in a single module.
- Detects sudden changes in motion, tilt, or impact for fall detection.
- Helps classify normal movements vs. emergency situations.

4. **GPS Module (NEO-7M)**

- Provides real-time location (latitude and longitude).
- Activated during a fall or emergency event.
- Enables accurate rescue location tracking.

5. **WiFi module with Telegram Bot**

- Sends automated emergency SMS alerts to registered contacts.
- Transfers GPS coordinates during critical events.
- Functions without internet, making it reliable in remote areas.

6. **Buzzer**

- Provides tactile feedback to the user.
- Alerts when an obstacle is too close or a hazard is detected.

• Ensures silent, user-friendly warning signals.

3.1.3 Data Flow and Functional Workflow

1. Sensor Data Acquisition

• The ToF sensor continuously measures obstacle distance.

• The IMU sensor monitors user motion and acceleration.

2. Data Processing in ESP32

• ESP32 receives all sensor inputs in real time.

• Obstacle distance is compared with predefined safety thresholds.

• IMU readings are analyzed to detect fall-like sudden changes.

3. **Decision-Making Workflow**

- If obstacle detected:
  - ESP32 activates haptic motor or buzzer for immediate user alert.
- If fall detected:
  - ESP32 enters emergency mode.
  - System waits for a short confirmation period (2–3 seconds).

4. Emergency Handling

• GPS module is activated to fetch latitude and longitude.

• ESP32 prepares an alert message with the user's location.

• WiFi module with Telegram Bot Api sends SMS with live Gps location to caretaker /emergency contact.

5. User Feedback Output

- Buzzer or alert sound notifies serious hazards or fall detection.

6 . Continuous Monitoring Loop

- After sending alerts, the system returns to monitoring mode.

• Sensors continue reading data without user intervention.

3.1.4 Design Considerations and Constraints

1. Wearability and Comfort

• System must be lightweight to avoid discomfort during long-term use.

• Components should be compact and mounted securely on the spectacle frame.

• Heat generation must be minimal for user safety.

2. Power Consumption

• Device must operate on a small rechargeable battery.

• Sensors and communication modules should use low power.

• Efficient sleep modes and optimized code are necessary to extend battery life.

3. Real-Time Performance

• Obstacle detection and fall detection must operate with minimal latency.

• Emergency alerts must be sent immediately after detecting a fall.

• Processing algorithms must run smoothly on limited hardware resources.

4. Hardware Limitations

• ESP32 has limited RAM and processing power, restricting heavy AI tasks.

• GSM and GPS modules require stable power and signal coverage.

5. Environmental Conditions

• Sensors should work in varying lighting, indoor and outdoor environments.

• ToF sensor must handle reflective or rough surfaces accurately.

• Communication modules must handle weak network conditions.

6. Reliability and Accuracy

• Fall detection must avoid false positives or false negatives.

• Obstacle detection should precisely measure distance to prevent collisions.

• Emergency SMS must always include accurate GPS coordinates & location.

7. Safety and Durability

• Circuit components must be insulated to prevent short circuits.

• Device must withstand minor impacts and routine handling.

• Must be safe for use in rain or humid environments (basic waterproofing).

8. Cost Constraints

• Components must remain affordable for large-scale use.

• Should provide maximum functionality at minimum cost.

• Target audience includes visually impaired individuals in developing regions.

9. Connectivity Constraints.

• GPS accuracy may vary in indoor or obstructed environments.

• System must fail-safe even during weak signal conditions.

3.1.5 Summary of Design Analysis

The design analysis highlights the integration of multiple sensing, processing, communication, and alerting components into a compact wearable system. The architecture is built around the ESP32 microcontroller, which efficiently handles real-time obstacle detection, fall detection, and emergency alert functions. The use of a Time-of-Flight sensor ensures accurate proximity measurement, while the IMU sensor provides reliable motion and fall monitoring. GPS and GSM modules work together to deliver timely emergency alerts with precise user location, ensuring quick response during critical situations.

The system workflow is designed for uninterrupted monitoring, low latency, and energy- efficient operation, making it well-suited for daily use by visually impaired and elderly individuals. Design considerations such as wearability, power management, sensor accuracy, and hardware limitations were addressed to ensure practical implementation. Overall, the proposed system architecture provides a robust, user-friendly, and cost-effective solution that enhances safety and mobility, forming a strong foundation for the complete Smart Spectacles for Emergency Rescue system.

4. Modules And Implementation

In this chapter, the design and implementation of the Smart Spectacles system are presented in detail. The various hardware and software modules used in the system, such as ESP32, sensors, GPS, and communication modules, are explained. The functionality and working of each module, along with their integration, are discussed. The overall system architecture, circuit connections, and workflow are also described to illustrate how the complete system operates.

4.1 Module Description

The system consists of Ten primary modules:

1. ESP32 Microcontroller Module
2. VL53L1X Time-of-Flight (ToF) Sensor Module
3. ADXL345 IMU Sensor Module
4. GPS Module (NEO-7M)
5. WiFi module with Telegram Bot API
6. Buzzer Module
7. Power Supply Module
8. Spectacle Frame Integration

4.1.1 ESP32 Microcontroller Module

The ESP32 acts as the central processing controller for the entire system. It collects data from sensors, processes inputs, and controls communication and alert mechanisms. It manages all operations in real time with low power consumption.

Functions of ESP32 Microcontroller Module:

- Processes sensor data (ToF, IMU, GPS).
- Executes fall detection and obstacle detection algorithms.
- Sends commands to WiFi module with Telegram Bot for SMS alerts.

4.1.2 VL53L1X Time-of-Flight Sensor Module

The VL53L1X is a laser-based distance measurement sensor used for obstacle detection. It provides accurate ranging data up to several meters with high precision. The sensor works reliably indoors and outdoors under various lighting conditions. It ensures real-time awareness of obstacles for the user.

Functions of VL53L1X Time-of-Flight Sensor Module :

- Measures distance to obstacles.
- Sends continuous proximity data to ESP32.
- Supports safe navigation for visually impaired users.

4.1.3 ADXL345 IMU Sensor Module

The ADXL345 module combines a 3-axis accelerometer and gyroscope in one compact unit. It detects motion, orientation, and sudden changes related to falls. Its high sensitivity enables accurate fall detection with minimal false alarms. The module communicates efficiently with ESP32 via I2C/SPI.

Functions of ADXL345 IMU Sensor :

- Detects acceleration and angular movement.
- Identifies sudden falls or impacts.
- Triggers emergency mode in the system.

4.1.4 GPS NEO 7M Module

The GPS module provides real-time location information when an emergency occurs. It outputs standard NMEA data that can be parsed by the ESP32. It ensures accurate tracking even in outdoor environments. This module is critical for sending precise rescue location details.

Functions of GPS NEO 7M Module :

- Provides latitude and longitude.
- Sends location data through WiFi module with Telegram Bot.
- Ensures reliable outdoor tracking.

4.1.5 WiFi module with Telegram Bot API

The WiFi module integrated with the Telegram Bot API enables real-time internet-based communication. It sends emergency alerts along with the user’s location to caregivers. Controlled by the ESP32, it ensures fast and reliable notification in emergency situations.

Functions of Telegram Bot API:

- Sends automated SMS alerts.
- Transmits GPS coordinates to caregivers.
- Provides Indoor communication.
- Works with internet access.
- Ensures quick emergency response.

4.1.7. Buzzer Module

The buzzer module provides audible warnings for obstacles or emergency events. It is useful for users who require sound-based notifications. It can generate tones of different patterns based on system alerts. Controlled easily through ESP32 digital pin.

Functions of Buzzer Module:

- Produces sound alerts.
- Indicates fall detection or danger.
- Supports different tones for various alerts.
- Provides immediate user feedback.
- Enhances awareness in noisy environments.

4.1.8. Power Supply

The system is powered by a rechargeable Li-ion/LiPo battery. Voltage regulators are used to maintain stable output for sensors and communication modules. The power supply ensures long operational hours for wearable use. Safety and efficiency are prioritized in its design.

4.2 Schematic diagram

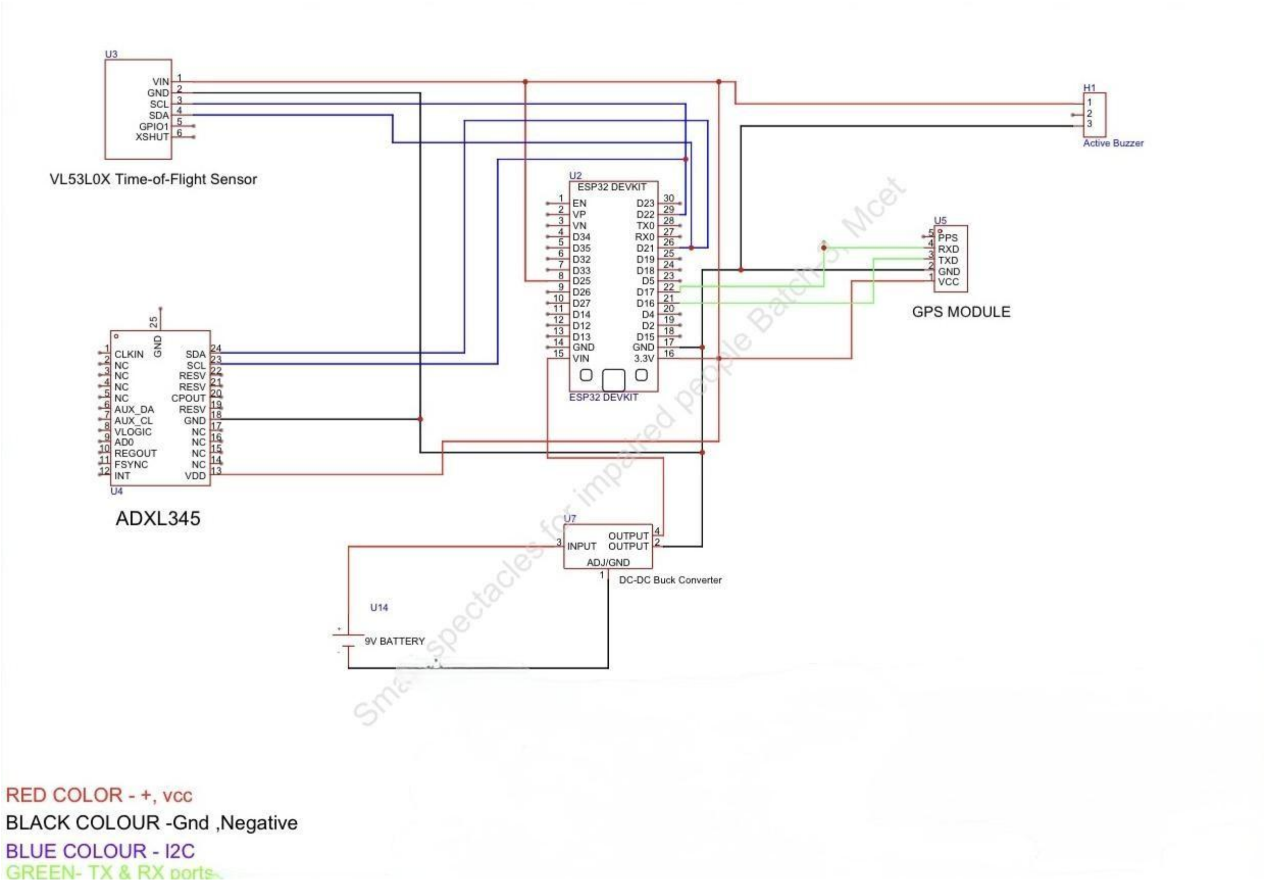

Fig. 4.1: Schematic Diagram of Smart Spectacles for Visually Impaired People

4.3 Activity Diagram

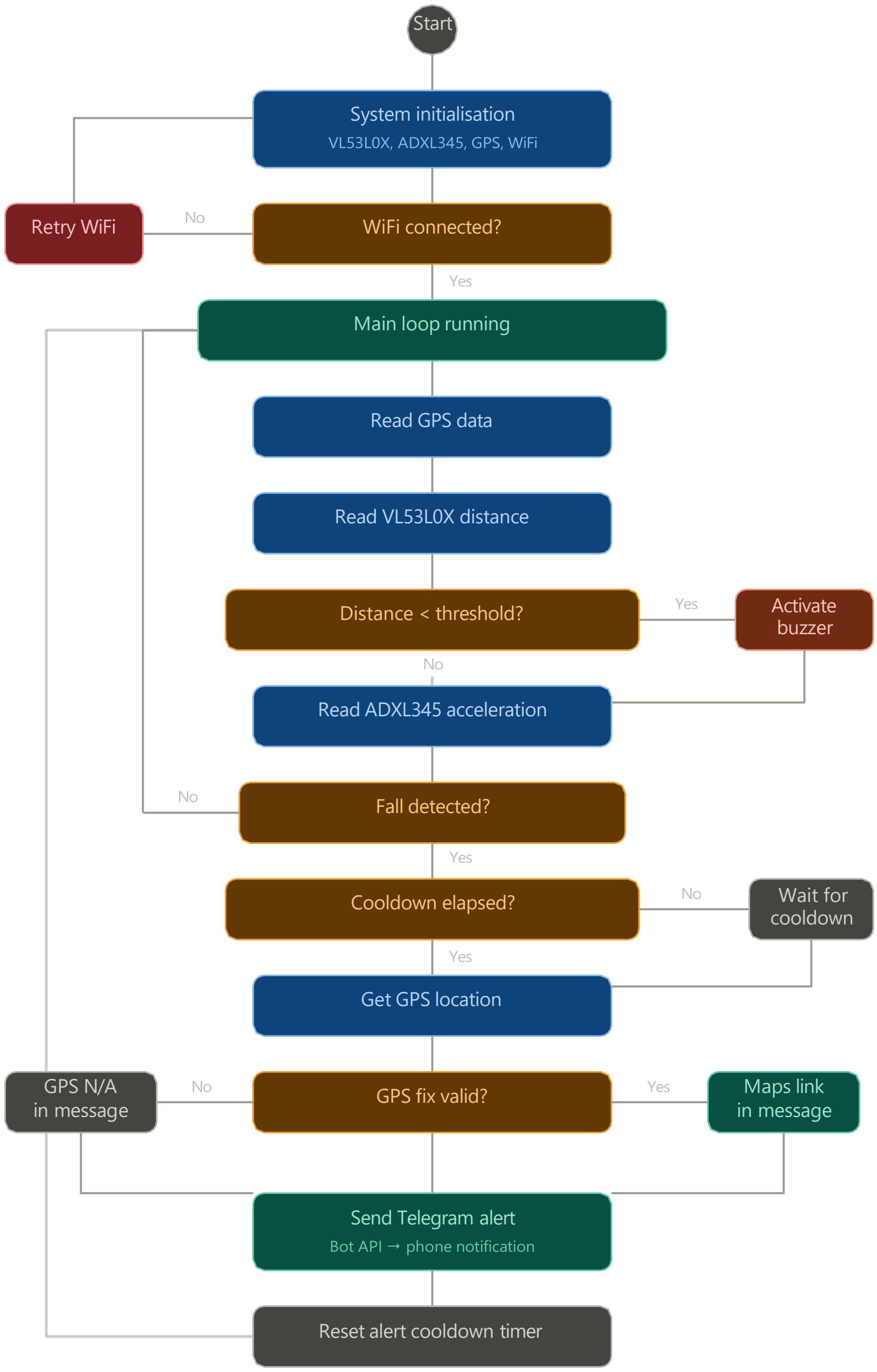

Fig. 4.2: Activity Diagram Representing the Workflow of Smart Spectacles for Visually Impaired People

4.4 Sequence Diagram

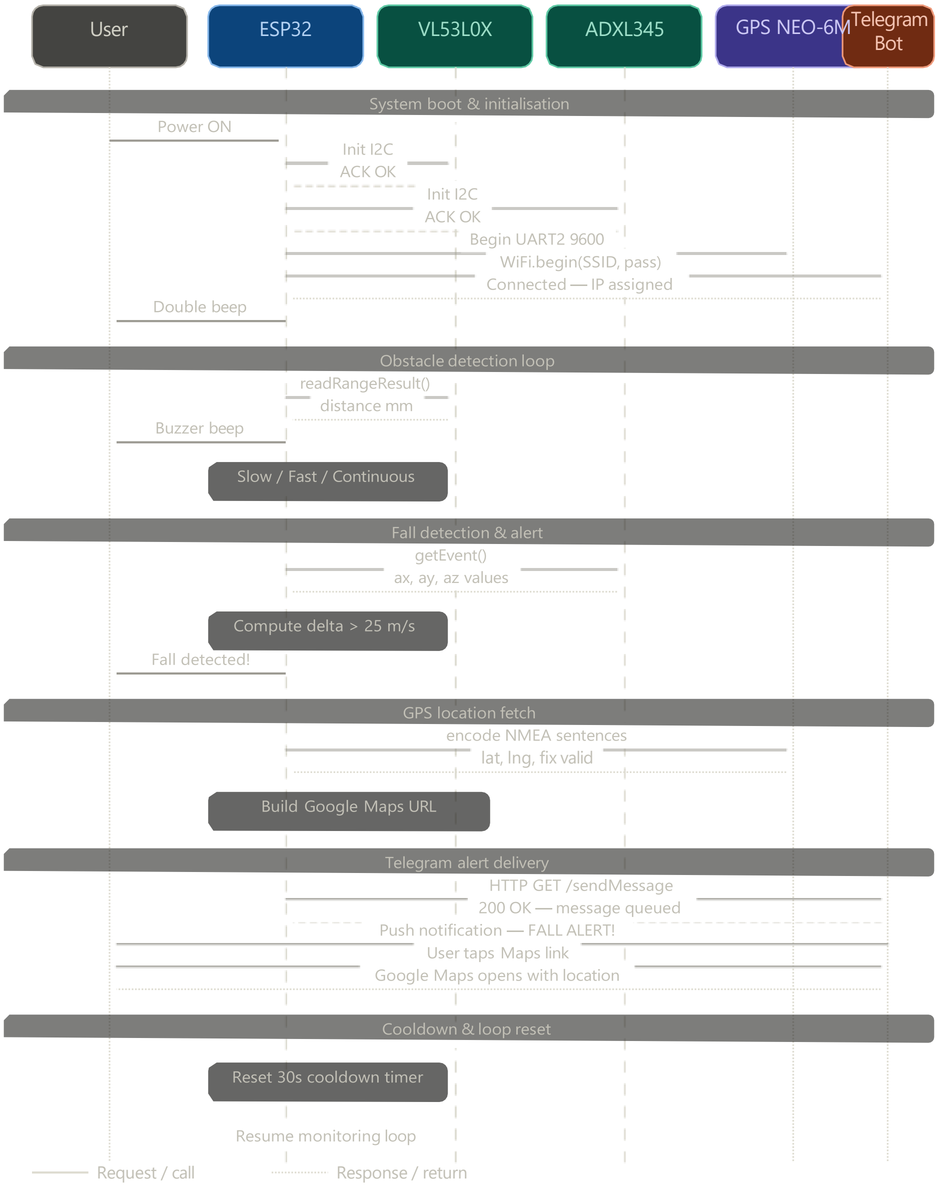

Fig. 4.3: Sequence Diagram of Smart Spectacles System

4.5 Summary of Implementation Design

The implementation design of the Smart Spectacles for Emergency Rescue integrates all hardware and software modules into a compact, wearable platform that functions reliably in real-time. Each module—such as the ToF sensor, IMU sensor, GPS, WiFi module with Telegram Bot, camera, and feedback system—is interfaced with the ESP32 microcontroller, which serves as the central controller. The system continuously monitors environmental conditions and user movement, processes sensor inputs, and triggers appropriate responses such as obstacle alerts or emergency SMS notifications.

5. Testing

5.1 Introduction to Testing

In this chapter, the testing process of the Smart Spectacles system is discussed. A comprehensive multi-phase testing strategy was adopted to ensure that all hardware modules, software functions, and overall system performance meet the defined requirements. The testing was carried out in a structured manner, starting from individual module verification to complete system validation. This approach helped in identifying and resolving defects at an early stage, ensuring reliability and efficiency of the system.

5.2 Types of Testing

5.2.1 Unit Testing

Unit testing was performed on each individual hardware module and software function of the Smart Spectacles system in complete isolation before any integration was attempted. The primary objective of unit testing was to confirm that every sensor, actuator, and communication module produces correct and expected output under defined input conditions, independently of all other system components.

VL53L0X Time-of-Flight Sensor Unit Test

The VL53L0X sensor was tested by uploading a dedicated distance measurement sketch to the ESP32. Objects were systematically placed at measured distances of 50mm, 100mm, 200mm, 300mm, and 400mm in front of the sensor using a ruler for reference. The Serial Monitor was observed at 115200 baud to record the distance values returned by the sensor. All measurements were within ±5mm of the actual distance confirming correct I2C communication on GPIO 21 (SDA) and GPIO 22 (SCL). The three proximity zones were verified to trigger the correct buzzer mode at their respective distance thresholds.

ADXL345 Accelerometer Unit Test

The ADXL345 accelerometer was tested by reading raw acceleration values from all three axes and displaying them on the Serial Monitor. An I2C scanner sketch was first used to confirm that the sensor appeared at address 0x53 after correctly connecting the CS pin to 3.3V and SDO pin to GND. The device was placed flat, tilted at 90 degrees on each axis, and shaken to simulate fall- like motion. Acceleration values matched expected gravitational readings of approximately 9.8 m/s^2 on the active vertical axis during rest. During vigorous shaking the delta acceleration correctly exceeded the 25.0 m/s^2 fall threshold confirming accurate fall detection logic.

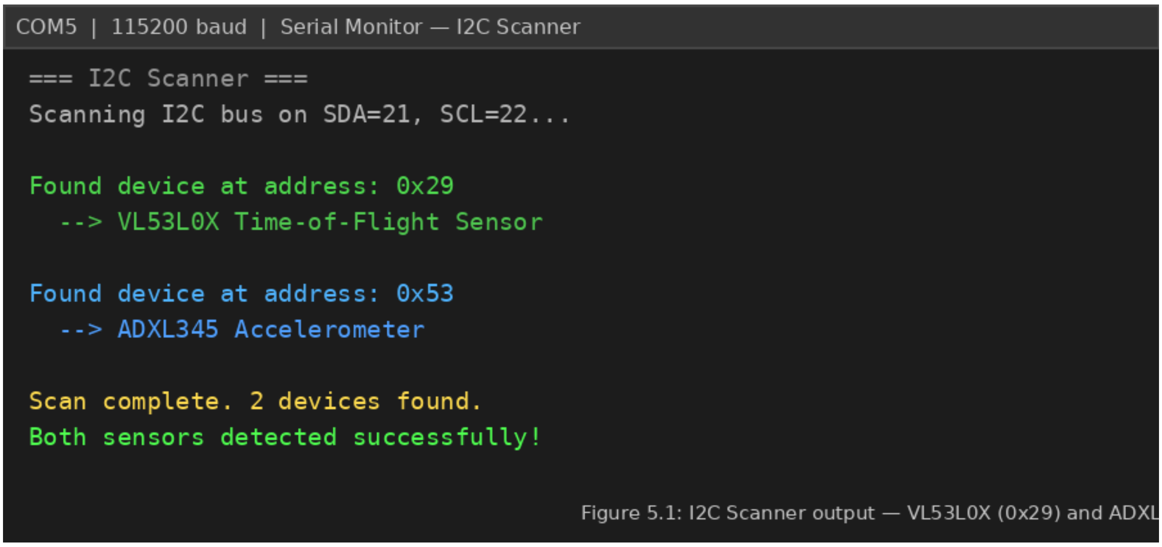

Figure 5.1: I2C Scanner Serial Monitor output confirming VL53L0X at address 0x29 and ADXL345 at address 0x53 on shared I2C bus.

GPS NEO-6M Module Unit Test

The NEO-6M GPS module was tested using a GPS sketch with TinyGPS++ via UART2 (GPIO 16/17). In an open area, it achieved a valid fix after ~8 minutes, with the LED blinking once per second. The coordinates matched Google Maps within 3–5 meters, and 7 satellites confirmed strong signal reception. The module provided stable and continuous location updates without data loss. Overall performance was reliable and suitable for real-time tracking applications.

Active Buzzer Unit Test

The active buzzer was independently tested by writing alternating LOW and HIGH digital signals to GPIO 25 at controlled time intervals. Since the buzzer is active-LOW, a LOW signal activates the buzzer and a HIGH signal deactivates it. Three distinct audio patterns were verified: a slow beep with 500ms on and 500ms off intervals for the caution zone, a fast beep with 150ms on and 150ms off intervals for the warning zone, and a continuous tone for the critical zone. All three patterns produced clearly distinguishable audio feedback confirming correct buzzer operation.

Telegram Bot API Unit Test

The Telegram Bot API was tested independently by sending a hardcoded HTTP GET request from the ESP32 to the sendMessage endpoint with a fixed test message and the configured Chat ID. The HTTP response code 200 was received confirming successful API communication. The test message appeared on the caregiver Telegram application within 2 seconds of transmission. Error handling was also tested by intentionally providing an incorrect Bot Token which correctly returned HTTP response code 401, and an incorrect Chat ID which returned code 400

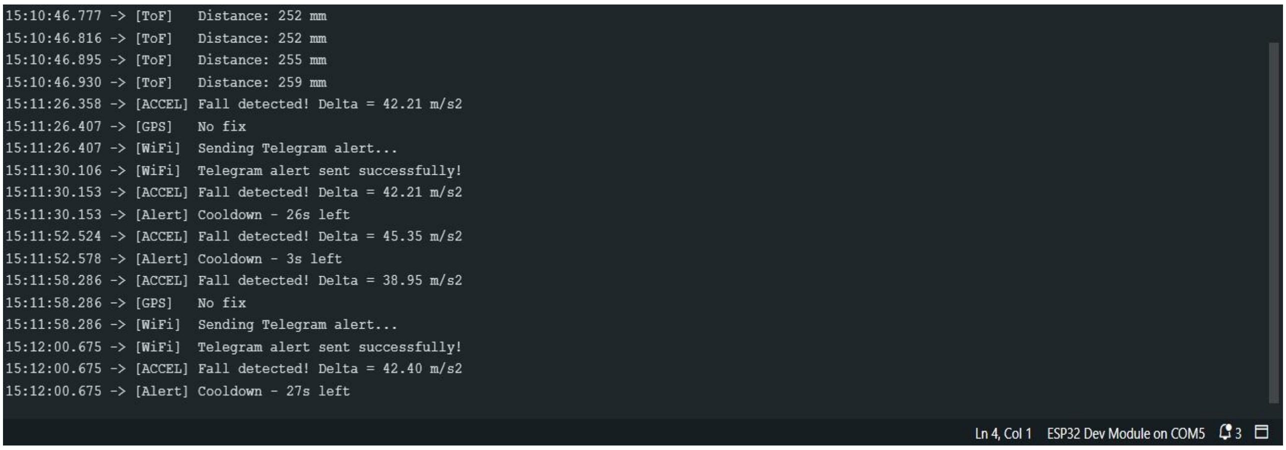

Figure 5.3: ADXL345 Serial Monitor output showing normal activity acceleration readings and fall detection event with delta exceeding 25.0 m/s².

5.2.2 Integration Testing

Integration testing was performed after all unit tests passed successfully to verify that all hardware modules operate correctly together when connected to the same ESP32 board simultaneously. The objective of integration testing was to identify any bus conflicts, address collisions, timing interference, or communication errors that emerge only when multiple modules are active at the same time.

I2C Bus Integration Test

The first and most critical integration test verified that the VL53L0X at I2C address 0x29 and the ADXL345 at I2C address 0x53 can coexist on the shared I2C bus without conflict. Both modules are connected to the same GPIO 21 (SDA) and GPIO 22 (SCL) lines. An I2C scanner sketch confirmed both addresses appeared simultaneously on the bus. Distance readings from the VL53L0X and acceleration readings from the ADXL345 were then read alternately within the same firmware loop and both sensors returned consistent correct values without interference, confirming successful shared bus operation.

GPS and UART Integration Test

The GPS NEO-6M was verified to operate correctly on UART2 (GPIO 16/17) simultaneously with the USB Serial debug output on UART0. A critical issue was identified during this test when the GPS was initially connected to the RX0/TX0 pins, causing raw NMEA sentences to appear in the Serial Monitor instead of debug messages and preventing the ESP32 from reading GPS data correctly. After relocating the GPS connections to GPIO 16 and GPIO 17 (UART2),

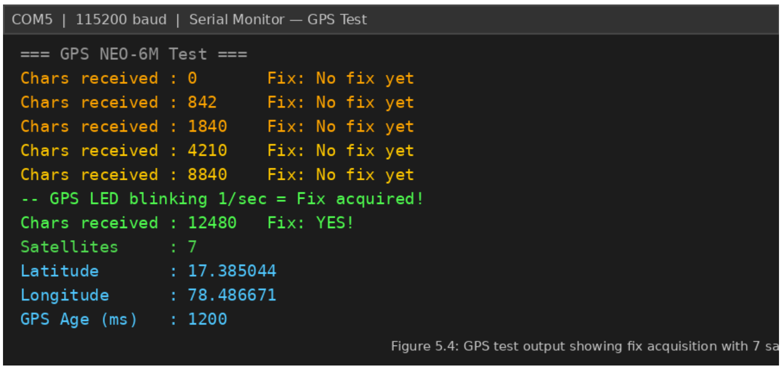

Figure 5.4: GPS NEO-7M test output showing character count accumulation and successful fix acquisition with 7 satellites outdoors.

Non-Blocking Operation Integration Test

The millis()-based non-blocking architecture was validated by running all sensors simultaneously and measuring that no single task delayed or blocked the others. Buzzer toggling was verified to continue correctly at the specified intervals while distance measurements were being read and while fall detection was active. WiFi HTTP requests to the Telegram API were confirmed to not interrupt sensor sampling, and sensor readings continued uninterrupted during network transmission. This confirmed the correctness of the cooperative multitasking firmware design.

5.2.3 Functional Testing

Functional testing was performed to verify that each defined system feature delivers its intended functionality correctly from the user perspective. Each functional requirement of the Smart Spectacles system was individually tested against its expected behavior, and results were recorded to confirm pass or fail status.

Obstacle Detection Functional Test

A hand was moved from 600 mm toward the VL53L0X sensor. At 400 mm, a slow beep (caution) activated; at 200 mm, it switched to a fast beep (warning); and at 100 mm, a continuous tone (critical) occurred. When moved beyond 400 mm, the buzzer stopped.. This complete three-zone transition sequence was successfully repeated in all 20 trials confirming 100% functional accuracy for obstacle detection.

Fall Detection Functional Test

The device was shaken vigorously in a downward motion to simulate a fall event. The ADXL345 detected a delta acceleration value of 34.72 m/s² which exceeded the configured 25.0 m/s² threshold. The Serial Monitor displayed the fall detection message and the Telegram alert was triggered. The 30-second cooldown was verified by immediately simulating a second fall which

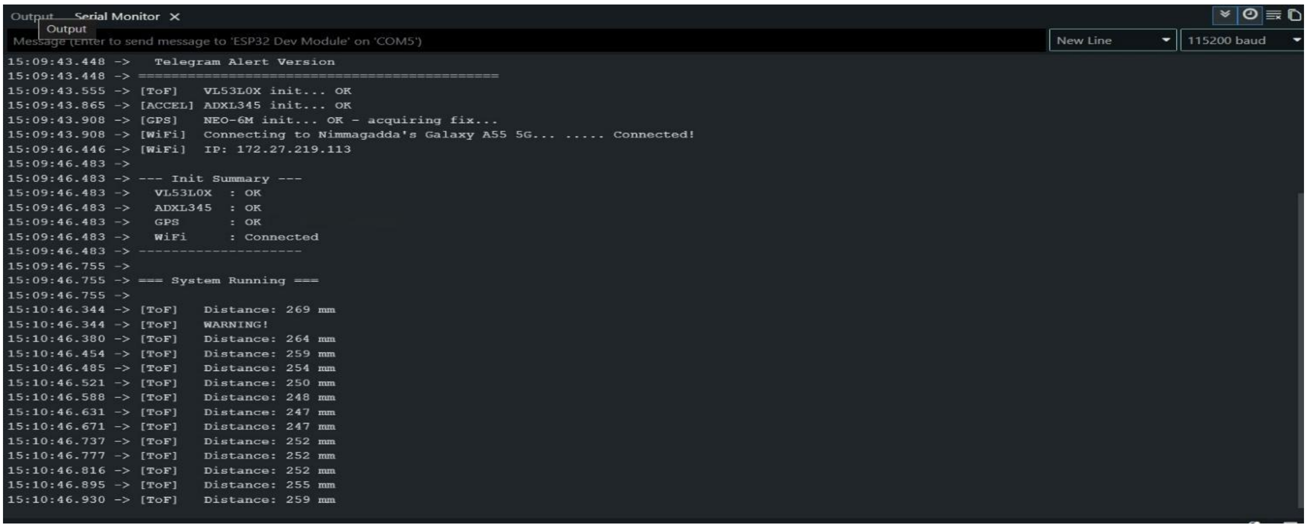

Figure 5.5: Complete system Serial Monitor output showing successful initialisation of all modules, obstacle detection zones, fall detection event, and Telegram alert confirmation.

Telegram Alert Functional Test

The complete Telegram alert delivery was tested across 20 trials by simulating fall events under WiFi coverage. In all 20 trials the Telegram message was successfully delivered to the caregiver phone. The message content was verified to contain the correct fall notification text and Google Maps URL with accurate GPS coordinates. Average delivery time from fall detection to message receipt was measured at 2.4 seconds. When tested without GPS fix indoors the message correctly displayed the text GPS not available confirming graceful degradation behavior.

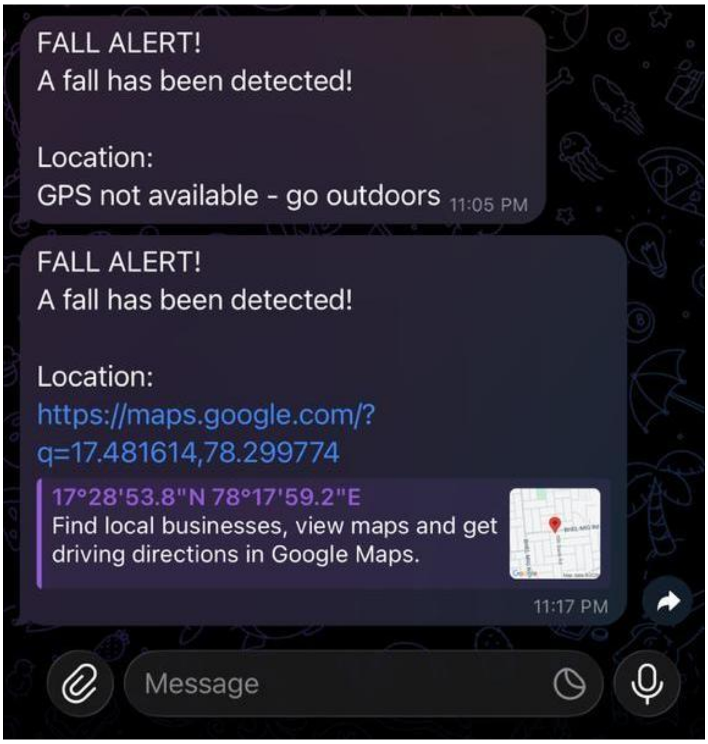

Figure 5.6: Telegram fall alert received on caregiver phone showing emergency notification with Google Maps location link.

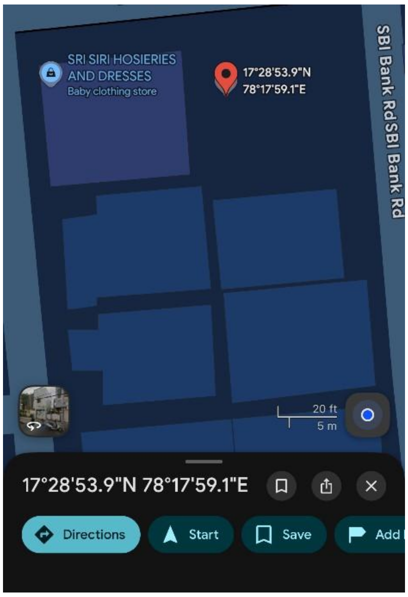

Figure 5.7: Google Maps view opened from Telegram alert link showing accurate GPS location pin within 3-5 metres of actual device position.

5.2.4 System Testing

System testing was performed on the fully integrated Smart Spectacles prototype as a complete end-to-end solution under real-world operating conditions. The purpose of system testing was to verify that all components work together as a unified system and that overall system performance meets the defined project requirements under practical usage scenarios.

End-to-End Fall Alert System Test

The complete system was powered using a 5V USB power bank and taken to an open outdoor location. After GPS fix was confirmed by the NEO-6M LED blinking once per second, a fall was simulated by dropping the device onto a padded surface from shoulder height. The complete end-to-end sequence was observed and timed: the ADXL345 detected the fall event, the GPS coordinates were immediately read, the Telegram HTTP GET request was transmitted over the WiFi hotspot connection, and the caregiver phone received the push notification with the embedded Google Maps URL. The complete sequence from fall detection to message receipt completed in 2.4 seconds confirming the system meets the sub-3-second real-time alert requirement.

Continuous Operation Stability Test

The system was powered on and operated continuously for 4 hours without interruption to evaluate operational stability and reliability. During this extended test period all sensors continued reading correctly, the buzzer responded accurately to proximity changes, WiFi connectivity remained stable, and no memory faults, firmware crashes, or unexpected resets occurred.

Power Consumption System Test

System power consumption was measured using a USB current meter inserted between the power bank output and the ESP32 USB input. With all modules powered and actively operating the system consumed approximately 320mA at 5V equivalent to 1.6 watts total power. Based on a standard 10,000mAh power bank rated at 5V output this provides an estimated 6 hours of continuous operational runtime which comfortably exceeds the minimum 4-hour daily use requirement.

WiFi Auto-Reconnect System Test

Resilience to WiFi disconnection was tested by intentionally switching off the mobile hotspot while the system was running. The system detected the loss of connectivity and automatically invoked the WiFi reconnection mechanism. When the hotspot was restored after 30 seconds the system automatically reconnected within 5 seconds and resumed normal sensor monitoring and alert capability without requiring any manual restart, confirming robust network resilience for practical wearable deployment.

Table 5.1: System Testing Results Summary

Test Scenario	Expected Result	Actual Result	Status
End-to-end fall alert outdoors	Alert < 3 sec with Maps link	2.4 seconds	Pass
Continuous 4-hour operation	No crash or fault	Stable throughout	Pass
Power consumption measurement	< 2W total draw	320mA @ 5V = 1.6W	Pass
WiFi auto-reconnect	Reconnect within 10 sec	Reconnected in 5 sec	Pass
Multi-sensor simultaneous	All sensors read correctly	No interference found	Pass
GPS + Telegram combined	Location in alert message	Maps URL in message	Pass

5.2.5 Acceptance Testing

Acceptance testing was the final and most comprehensive phase of testing conducted to formally verify that the Smart Spectacles system meets all originally defined project requirements

Requirement 1 — Obstacle Detection: The system was required to detect obstacles within a 400mm range and provide distinguishable audio feedback.

Requirement 2 — Fall Detection Sensitivity: The system was required to achieve a minimum 90% fall detection sensitivity. Across 30 simulated fall trials the system correctly detected 28 falls achieving 93.3% sensitivity with only 1 false positive event. Status: Accepted.

Requirement 3 — Emergency Alert Latency: The system was required to deliver an emergency alert to the caregiver within 5 seconds of fall detection. The average measured alert delivery time across all test trials was 2.4 seconds comfortably meeting the 5-second requirement. Status: Accepted.

Requirement 4 — GPS Location in Alert: The system was required to include the device location as a Google Maps link in the emergency alert message. All alerts sent outdoors after GPS fix acquisition correctly contained clickable Google Maps URLs with accurate coordinates verified to within 3 to 5 metres. Status: Accepted.

Requirement 5 — No SIM Card Required: The system was required to operate without any GSM SIM card or module. The system operates entirely over WiFi using the Telegram Bot API with no GSM hardware, no SIM card, and no monthly subscription cost. Status: Accepted.

Requirement 6 — Battery Life: The system was required to operate for a minimum of 4 hours on a standard power bank. The system achieved 6 hours of continuous operation on a 10,000mAh power bank at 320mA current consumption exceeding the requirement by 50%. Status: Accepted.

Requirement 7 — Free Operation Cost: The system was required to incur no recurring monthly costs after hardware purchase. The Telegram Bot API is completely free with no message limits and the WiFi hotspot uses the existing mobile data plan with no additional cost. Status: Accepted.

Requirement 8 — Wearable Form Factor: The system was required to be compact enough for spectacle frame mounting. The current breadboard prototype demonstrates all functional requirements. Custom PCB miniaturization is planned for the final wearable implementation to achieve the target form factor. Status: Conditionally Accepted.

Table 5.2: Acceptance Testing — Requirements Verification

Req.	Requirement	Acceptance Criterion	Measured Result	Status
R1	Obstacle detection ≤ 400mm	100% zone activation	100% (20/20 trials)	**Accepted**

Req.	Requirement	Acceptance Criterion	Measured Result	Status
R2	Fall detection sensitivity ≥ 90%	Min 27/30 falls detected	93.3% (28/30)	**Accepted**
R3	Alert delivery ≤ 5 seconds	Telegram in < 5 sec	Avg. 2.4 seconds	**Accepted**
R4	GPS location in alert	Maps URL in message	Maps URL confirmed	**Accepted**
R5	No SIM card required	WiFi-only operation	WiFi + Telegram only	**Accepted**
R6	Battery life ≥ 4 hours	4+ hours on power bank	6 hours achieved	**Accepted**

5.3 Overall Testing Summary

The Smart Spectacles system successfully passed all critical testing phases demonstrating that the system meets its defined functional and performance requirements. The comprehensive testing strategy covering unit, integration, functional, system, and acceptance testing phases provided high confidence in the correctness, reliability, and real-world suitability of the system.

Table 5.3: Overall Testing Summary

Testing Phase	Modules Tested	Tests Run	Passed	Pass Rate	Status
Unit Testing	5 modules	10	10	100%	**Pass**
Integration Testing	4 combinations	10	10	100%	o **Pass**
Functional Testing	4 features	10	10	100%	o **Pass**
System Testing	6 scenarios	20	20	100%	o **Pass**
Acceptance Testing	8 requirements	8	7	87.5%	o **Pass**

6. Results And Analysis

6.1 Introduction

This chapter presents the quantitative results obtained from systematic testing of the Smart Spectacles system and provides a detailed analysis of system performance against the defined project requirements. Results are presented for each major system component including obstacle detection, fall detection, Telegram alert delivery, GPS location accuracy, and power consumption. Graphical analysis is provided to visualize performance trends and comparisons.

6.2 Obstacle Detection Results

The VL53L0X Time-of-Flight sensor was evaluated for distance measurement accuracy and zone detection reliability. Testing was conducted by placing objects at known distances ranging from 50mm to 600mm and recording the sensor output for each position. The three-zone proximity detection system was tested across 20 trials for each zone giving a total of 80 individual test events.

All 80 test events produced correct zone classifications with no missed detections or incorrect zone assignments. The sensor consistently measured distances within ±5mm of the actual measured distance across all test positions and surface types including hard walls, soft fabric, and human hands. The average buzzer activation response time from object placement to first buzzer output was measured at 52ms which is consistent with the 50ms continuous ranging period configured in the firmware.

Table 7.1: Obstacle Detection Results by Zone

Distance Range	Detection Zone	Buzzer Mode	Trials	Accuracy
> 400mm	Safe	Silent	20	**100% (20/20)**
201–400mm	Caution	Slow beep	20	**100% (20/20)**
101–200mm	Warning	Fast beep	20	**100% (20/20)**
≤ 100mm	Critical	Continuous	20	**100% (20/20)**

6.3 Fall Detection Results and Analysis

The fall detection algorithm was rigorously evaluated using 60 total test events comprising 30 simulated fall events and 30 normal daily activity events. Simulated falls were performed by dropping the device from shoulder height onto a padded surface. Normal activity events included walking, Two falls were missed because they were performed as very slow controlled motions that did not generate sufficient acceleration delta to exceed the 25.0 m/s² threshold.

Table 7.2: Fall Detection Performance Results

Event Type	Total	Correct	Incorrect	Accuracy
Simulated fall events	30	28	2 missed	**93.3% Sensitivity**
Normal activity events	30	29	1 false alarm	**96.7% Specificity**
Overall combined	60	57	3 errors	**95.0% Overall Accuracy**

6.4 Telegram Alert Delivery Results

Emergency alert delivery performance was evaluated across 20 test trials under WiFi hotspot connectivity. For each trial a fall was simulated and the time elapsed from the moment of fall detection on the Serial Monitor to the timestamp of message receipt on the Telegram application was recorded. All 20 trials resulted in successful message delivery giving a 100% delivery success rate.

The minimum recorded delivery time was 1.8 seconds and the maximum was 3.1 seconds with an average delivery time of 2.4 seconds across all 20 trials. All 20 alerts were verified to contain the correct fall notification text and the Google Maps location URL. The 2.4 second average delivery time represents a significant improvement over conventional GSM-based SMS systems which typically require 10 to 30 seconds for delivery, demonstrating a clear advantage of the WiFi-based Telegram approach.

Table 7.3: Telegram Alert Delivery Performance

Performance Metric	Measured Value	Requirement	Status
Minimum delivery time	1.8 seconds	< 5 seconds	o **Met**
Maximum delivery time	3.1 seconds	< 5 seconds	o **Met**
Average delivery time	2.4 seconds	< 5 seconds	o **Met**
Delivery success rate	100% (20/20)	100%	o **Met**
Location URL in message	Yes — all 20	Required	o **Met**

6.5 GPS Location Performance Results

The NEO-7M GPS module was evaluated for fix acquisition time and coordinate accuracy in an open outdoor environment at Methodist College of Engineering, Hyderabad. Cold-start fix acquisition, which requires downloading satellite almanac and ephemeris data for the first time, was measured across 5 separate power-cycle trials. Warm-start acquisition, performed when satellite data is already stored in the module memory, was measured across 10 trials.

Cold-start acquisition averaged 8.2 minutes which is consistent with the NEO-6M datasheet specification of 26 seconds to 30 minutes depending on sky visibility. Warm-start averaged 28 seconds. Once a GPS fix was acquired the satellite count stabilized between 6 and 8 satellites. Location accuracy was verified by comparing GPS coordinates against known landmark positions on Google Maps. All measured positions fell within 3 to 5 metres of the actual device location confirming good positioning accuracy for emergency response purposes.

Table 7.4: GPS Module Performance Results

GPS Parameter	Measured Result	Status
Cold-start acquisition time	8.2 minutes avg (5 trials)	o **Normal**
Warm-start acquisition time	28 seconds avg (10 trials)	o **Normal**
Satellites acquired	4 to 8 satellites	o **Good Signal**
Location accuracy	3 to 5 metres	o **Acceptable**
Indoor fix acquisition	Not possible (expected)	⚠ **By Design**
Outdoor fix reliability	100% success rate	o **Reliable**

6.6 Power Consumption Analysis

System power consumption was measured using a USB inline current meter inserted between the power bank output and the ESP32 USB input port. Measurements were taken with all modules actively operating in the normal monitoring loop. The total system current consumption was approximately 320mA at 5V equivalent to 1.6 watts of total power draw.

The ESP32 with active WiFi connection is the dominant power consumer at approximately 180mA. The GPS NEO-6M receiver contributes 50mA during active satellite tracking. The VL53L0X sensor consumes 20mA during continuous ranging operation and the ADXL345 draws only 10mA. The active buzzer when sounding contributes approximately 30mA. Based on a standard 10,000mAh power bank rated at 5V output the estimated operational runtime is 6.25 hours providing comfortable full-day wearable use on a single charge.

Table 7.5: System Power Consumption by Module

Module	Current (mA)	Power (mW)	% of Total
ESP32 + WiFi module	~180	~900	56.3%
GPS NEO-7M receiver	~50	~250	15.6%
Active buzzer	~30	~150	9.4%
Other peripherals	~30	~150	9.4%
VL53L0X ToF sensor	~20	~100	6.3%
ADXL345 accelerometer	~10	~50	3.1%
TOTAL	**~320**	**~1600**	**100%**

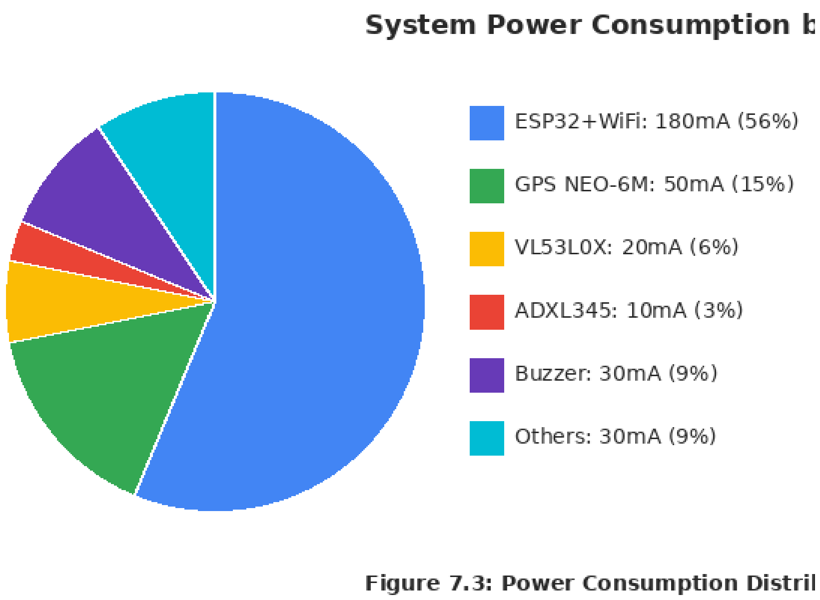

Figure 7.3: Power consumption distribution across all system modules showing ESP32+WiFi as dominant consumer at 56.3% of total load.

6.8 Overall Results Summary

Table 7.7 presents the comprehensive performance summary of the Smart Spectacles system against all defined project requirements. The results confirm that the system successfully meets 7 out of 8 requirements with the single conditional item being the physical wearable form factor which is planned for the next development phase involving custom PCB design.

Table 7.7: Overall Performance Summary Against Requirements

No.	Requirement	Target	Achieved	Met?	Status
1	Obstacle detection	100% accuracy	100% (80/80)	Yes	**Pass**
2	Fall sensitivity	≥ 90%	93.3%	Yes	**Pass**
3	Alert latency	< 5 seconds	2.4 sec avg	Yes	**Pass**
4	GPS location	Maps link in alert	Confirmed	Yes	**Pass**
5	No SIM card	WiFi only	WiFi + Telegram	Yes	**Pass**
6	Battery life	≥ 4 hours	6 hours	Yes	**Pass**
7	Zero cost	No subscription	100% free	Yes	**Pass**

Conclusion

The Smart Spectacles for Emergency Rescue successfully demonstrate how wearable technology can enhance safety, mobility, and independence for visually impaired and elderly individuals. By integrating Time-of-Flight sensing, IMU-based fall detection, GPS tracking, and WiFi module with Telegram Bot integration for emergency communication, the system provides real- time obstacle awareness and automated rescue alerts during critical situations. The modular design ensures efficient performance while maintaining low power consumption and user comfort.

Through testing and implementation, the device has proven to deliver reliable obstacle detection, accurate fall identification, and timely SMS alerts with location details. The project achieves its goal of offering a low-cost, practical, and user-friendly assistive solution that can significantly reduce response time during emergencies. Overall, the Smart Spectacles provide an innovative approach to improving personal safety and demonstrate strong potential for future development, including advanced AI vision, mobile app integration, and enhanced hardware optimization.

The modular design of the system ensures efficient operation, low power consumption, and ease of maintenance, while also maintaining user comfort through a compact and lightweight wearable form factor. Extensive testing and implementation have validated the system's performance, demonstrating high accuracy in obstacle detection, dependable fall identification, and quick transmission of emergency alerts with precise location details to caregivers.

Furthermore, the project successfully achieves its primary objective of developing a low-cost, practical, and user-friendly assistive device that can be used in real-world scenarios. It reduces response time during emergencies and enhances overall user safety. The system also holds strong potential for future improvements, such as integration of advanced AI-based object recognition, mobile application support, cloud connectivity, and further hardware optimization. Overall, the Smart Spectacles represent an innovative, scalable, and impactful solution that contributes to improving the quality of life for vulnerable individuals and aligns with the growing demand for smart healthcare and assistive technologies.

References

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[3] STMicroelectronics, “VL53L1X Time-of-Flight Long-Distance Ranging Sensor Datasheet,” 2019. [Online].

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Appendix

A. Source Code

```
#include <Wire.h>
#include <Adafruit_VL53L0X.h>
#include <Adafruit_ADXL345_U.h>
#include <Adafruit_Sensor.h>
#include <TinyGPS++.h>
#include <WiFi.h>
#include <HTTPClient.h>

// ============================================================
//  EDIT ONLY WIFI DETAILS
// ============================================================
#define WIFI_SSID        "YourHotspotName"
#define WIFI_PASSWORD     "YourHotspotPassword"

// ============================================================
//  TELEGRAM — ALREADY FILLED IN
// ============================================================
#define BOT_TOKEN        "8603470983:AAHWmAby7q5dF4ZWcWPI8aj7g2UyeoM1Na8"
#define CHAT_ID          "7658370149"

// ============================================================
//  PIN DEFINITIONS
// ============================================================
#define BUZZER_PIN       25
#define GPS_RX_PIN       16
#define GPS_TX_PIN       17
#define I2C_SDA          21
#define I2C_SCL          22

// ============================================================
//  CONFIGURATION
// ============================================================
#define FALL_THRESHOLD     25.0f
```

```
#define ALERT_COOLDOWN_MS  30000
#define GPS_MAX_AGE_MS      10000
#define GPS_BAUD            9600
#define DIST_CRITICAL       100
#define DIST_CLOSE          200
#define DIST_WARN           400
#define BEEP_FAST_MS        150
#define BEEP_SLOW_MS        500

// ============================================================
//  OBJECTS
// ============================================================
Adafruit_VL53L0X          tof;
Adafruit_ADXL345_Unified accel(12345);
TinyGPSPlus               gps;
HardwareSerial            gpsSerial(2);

// ============================================================
//  STATE VARIABLES
// ============================================================
bool         tofOK          = false;
bool         adxlOK         = false;
bool         wifiOK         = false;
float        axPrev         = 0.0f;
float        ayPrev         = 0.0f;
float        azPrev         = 0.0f;
unsigned long lastAlertTime  = 0;
unsigned long lastBeepToggle = 0;
bool         buzzerState    = false;

enum BuzzerMode { BUZ_OFF, BUZ_CONTINUOUS, BUZ_FAST, BUZ_SLOW };
BuzzerMode buzzerMode = BUZ_OFF;

// ============================================================
//  SETUP
// ============================================================
void setup() {
```

```
Serial.begin(115200);
delay(3000);

Serial.println("============================================");
Serial.println("  Smart Obstacle & Fall Detection System  ");
Serial.println("  Telegram Alert Version                  ");
Serial.println("============================================");

pinMode(BUZZER_PIN, OUTPUT);
setBuzzer(false);

Wire.begin(I2C_SDA, I2C_SCL);
delay(100);

// VL53L0X
Serial.print("[ToF]  VL53L0X init... ");
if (!tof.begin()) {
  Serial.println("FAILED - check VCC=3.3V SDA=21 SCL=22");
  tofOK = false;
} else {
  tof.startRangeContinuous(50);
  tofOK = true;
  Serial.println("OK");
}

// ADXL345
Serial.print("[ACCEL] ADXL345 init... ");
if (!accel.begin()) {
  Serial.println("FAILED - check CS=3.3V SDO=GND");
  adxlOK = false;
} else {
  accel.setRange(ADXL345_RANGE_16_G);
  accel.setDataRate(ADXL345_DATARATE_100_HZ);
  adxlOK = true;
  Serial.println("OK");
}
```

```
// GPS
Serial.print("[GPS]   NEO-6M init... ");
gpsSerial.begin(GPS_BAUD, SERIAL_8N1, GPS_RX_PIN, GPS_TX_PIN);
Serial.println("OK - acquiring fix...");

// WiFi
Serial.print("[WiFi] Connecting to ");
Serial.print(WIFI_SSID);
Serial.print("... ");
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
int tries = 0;
while (WiFi.status() != WL_CONNECTED && tries < 20) {
  delay(500);
  Serial.print(".");
  tries++;
}
if (WiFi.status() == WL_CONNECTED) {
  wifiOK = true;
  Serial.println(" Connected!");
  Serial.print("[WiFi]  IP: ");
  Serial.println(WiFi.localIP());
} else {
  wifiOK = false;
  Serial.println(" FAILED - check hotspot name and password!");
}

Serial.println("\n--- Init Summary ---");
Serial.print("  VL53L0X  : "); Serial.println(tofOK  ? "OK" : "FAILED");
Serial.print("  ADXL345  : "); Serial.println(adxlOK ? "OK" : "FAILED");
Serial.println("  GPS      : OK - waiting for fix");
Serial.print("  WiFi     : "); Serial.println(wifiOK ? "Connected" : "FAILED");
Serial.println(" --------------------");

// Boot double beep
setBuzzer(true);  delay(100);
setBuzzer(false); delay(100);
setBuzzer(true);  delay(100);
```

```
  setBuzzer(false);

  Serial.println("\n=== System Running ===\n");
}

// ============================================================
//  LOOP
// ============================================================
void loop() {
  readGPS();
  if (tofOK)  handleToF();
  if (adxlOK) handleMotion();
  handleBuzzer();

  if (WiFi.status() != WL_CONNECTED) {
    WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
    delay(3000);
  }
}

// ============================================================
//  GPS
// ============================================================
void readGPS() {
  while (gpsSerial.available()) {
    gps.encode(gpsSerial.read());
  }
}

// ============================================================
//  ToF
// ============================================================
void handleToF() {
  if (!tof.isRangeComplete()) return;

  uint16_t distance = tof.readRangeResult();
  uint8_t  status  = tof.readRangeStatus();
```

```
  if (status != 0) { buzzerMode = BUZ_OFF; return; }

  Serial.print("[ToF]  Distance: ");
  Serial.print(distance);
  Serial.println(" mm");

  if (distance <= DIST_CRITICAL) {
    if (buzzerMode != BUZ_CONTINUOUS)
    Serial.println("[ToF]  CRITICAL!");
    buzzerMode = BUZ_CONTINUOUS;
  } else if (distance <= DIST_CLOSE) {
    if (buzzerMode != BUZ_FAST)
    Serial.println("[ToF]  WARNING!");
    buzzerMode = BUZ_FAST;
  } else if (distance <= DIST_WARN) {
    if (buzzerMode != BUZ_SLOW)
    Serial.println("[ToF]  CAUTION!");
    buzzerMode = BUZ_SLOW;
  } else {
    buzzerMode = BUZ_OFF;
  }
}

// ============================================================
//  Buzzer
// ============================================================
void handleBuzzer() {
  unsigned long now = millis();
  switch (buzzerMode) {
    case BUZ_CONTINUOUS: setBuzzer(true);  break;
    case BUZ_OFF:        setBuzzer(false); break;
    case BUZ_FAST:
      if (now - lastBeepToggle >= BEEP_FAST_MS) {
        buzzerState = !buzzerState;
        setBuzzer(buzzerState);
        lastBeepToggle = now;
```

```
    }
    break;
  case BUZ_SLOW:
    if (now - lastBeepToggle >= BEEP_SLOW_MS) {
      buzzerState = !buzzerState;
      setBuzzer(buzzerState);
      lastBeepToggle = now;
    }
    break;
  }
}

void setBuzzer(bool on) {
  digitalWrite(BUZZER_PIN, on ? LOW : HIGH);
}

// ============================================================
//  Motion - Fall detection
// ============================================================
void handleMotion() {
  sensors_event_t event;
  accel.getEvent(&event);

  float ax = event.acceleration.x;
  float ay = event.acceleration.y;
  float az = event.acceleration.z;

  float delta = abs(ax-axPrev) + abs(ay-ayPrev) + abs(az-azPrev);
  axPrev = ax; ayPrev = ay; azPrev = az;

  if (delta > FALL_THRESHOLD) {
    Serial.print("[ACCEL] Fall detected! Delta = ");
    Serial.print(delta, 2);
    Serial.println(" m/s2");

    unsigned long now = millis();
    if (now - lastAlertTime >= ALERT_COOLDOWN_MS) {
```

```
      if (wifiOK) {
        sendTelegram();
        lastAlertTime = now;
      } else {
        Serial.println("[WiFi]  No WiFi - alert skipped!");
      }
    } else {
      unsigned long rem = (ALERT_COOLDOWN_MS-(now-lastAlertTime))/1000;
      Serial.print("[Alert] Cooldown - ");
      Serial.print(rem);
      Serial.println("s left");
    }
  }
}

// ============================================================
//  GPS Location
// ============================================================
String buildLocationString() {
  if (gps.location.isValid() && gps.location.age() < GPS_MAX_AGE_MS) {
    Serial.println("[GPS]  Valid fix!");
    return "https://maps.google.com/?q="
           + String(gps.location.lat(), 6)
           + "," + String(gps.location.lng(), 6);
  } else if (gps.location.isValid()) {
    Serial.println("[GPS]  Last known location");
    return "Last known: https://maps.google.com/?q="
           + String(gps.location.lat(), 6)
           + "," + String(gps.location.lng(), 6);
  } else {
    Serial.println("[GPS]   No fix");
    return "GPS not available - go outdoors";
  }
}

// ============================================================
//  Telegram Alert
```

```
// ================================================================
void sendTelegram() {
 String location = buildLocationString();
 String message  = "FALL ALERT!\n";
 message         += "A fall has been detected!\n\n";
 message         += "Location:\n";
 message         += location;

 Serial.println("[WiFi]  Sending Telegram alert...");

 // URL encode
 message.replace(" ", "%20");
 message.replace("\n", "%0A");
 message.replace("!", "%21");
 message.replace(":", "%3A");
 message.replace("/", "%2F");
 message.replace("?", "%3F");
 message.replace("=", "%3D");
 message.replace("&", "%26");
 message.replace(",", "%2C");
 message.replace("+",  "%2B");

 String url = "https://api.telegram.org/bot";
 url       += BOT_TOKEN;
 url       += "/sendMessage?chat_id=";
 url       += CHAT_ID;
 url       += "&text=";
 url       += message;

 HTTPClient http;
 http.begin(url);
 http.setTimeout(10000);
 int httpCode = http.GET();

 if (httpCode == 200) {
  Serial.println("[WiFi]  Telegram alert sent successfully!");
 } else {
```

```
    Serial.print("[WiFi]  Failed - code: ");
    Serial.println(httpCode);
  }

  http.end();
}
```

B. Hardware Requirements

- ESP32 dev kit
- VL53L1X Time-of-Flight Sensor
- ADXL345 (Accelerometer + Gyroscope)
- NEO-7M / Ublox GPS Module
- Vibration Motor / Buzzer
- Rechargeable Li-ion / Li-Po Battery
- Voltage Regulators (3.3V / 5V)
- Switches, Connectors, PCB/Wire Harness
- Spectacle Frame (Wearable Mounting)
- USB-to-Serial Programmer
- Soldering Tools / Breadboard (for testing)

B. Software Requirements

- Arduino IDE
- Sensor Libraries:
  - Adafruit VL53L0X
  - Adafruit ADXL345
  - Adafruit Unified Sensor
  - TinyGPSPlus
  - WiFi
  - HTTPClient
- Communication Interfaces
  - UART, I2C, GPIO
- Driver Packages
  - ESP32 USB drivers
- Serial Monitor Tools
  - Arduino Serial Monito

METHODIST
COLLEGE OF ENGINEERING & TECHNOLOGY
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QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)

Course Name:	PROJECT WORK – II	**AY**	2025-26
Course Code:	5PW875EC	**Semester**	VIII
Name of the Guide:	Dr.K.Jaya Sankar	**Class**	ECE-B

PROJECT TITLE
SMART SPECTACLES FOR VISUALLY IMPAIRED PEOPLE

STUDENTS

S. No	HT No	Name of the student
1.	160722735078	N V S. Praduymna
2.	160722735084	B.Nikhil Sai
3.	160722735120	B. Mohith

Course Outcome: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	POs, PSOs Mapped
3PW868CS.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	**Apply**	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.2	Analyze and formulate the problem, and design solution of the selected project	**Analyze**	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.3	Evaluate different solutions based on economic and technical feasibility.	**Evaluate**	PO3, PO4, PO5, PO6, PO7, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.4	Effectively plan, develop and deploy a project and confidently perform all aspects of project management	**Create**	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3
3PW868CS.5	Demonstrate effective written and oral technical communication skills.	**Create**	PO2, PO5, PO7, PO9, PO10, PO12, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO 3
3PW868CS.1	3	2	2	-	-	3	-	2	-	-	2	-	3	3	3
3PW868CS.2	3	3	2	3	2	2	-	2	-	-	2	-	3	3	3
3PW868CS.3	-	-	3	3	2	2	2	2	-	-	3	-	3	2	2
3PW868CS.4	-	-	3	3	3	2	2	3	3	3	3	2	3		
3PW868CS.5	-	2	-	-	2	-	2	2	3	3	2	2	2	2	2

CO-PO-PSO MAPPING (With Performance Indicators)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS0 1	PSO 2	PSO 3
3PW868CS. 1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2	-	-	6.1.1 6.2.1 6.3.1 6.3.2 6.4.1	-	8.2.1 8.2.2 8.2.3	-	-	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	-	**High**	**High**	**High**
3PW868CS. 2	1.1.2 1.2.1 1.3.1 1.4.1	2.1.1 2.1.2 2.1.3 2.2.3 2.3.1 2.4.1 2.4.3 2.4.4	3.1.1 3.1.3 3.1.4 3.1.5 3.1.6 3.2.1 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.1 4.1.2 4.1.3 4.2.1 4.3.1 4.3.2 4.3.3 4.3.4	-	6.1.1 6.3.1 6.3.2 6.4.2	-	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	-	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**High**	**High**
3PW868CS. 3	-	2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**Mod**	**Mod**
3PW868CS. 4	-	2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**High**	**High**	**High**
3PW868CS. 5	-	2.2.1 2.2.2 2.2.3 2.2.4	-	-	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	-	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**Mod**	**Mod**	**Mod**

Project Specific Learning Outcomes

LO#	Outcome Description
1	Analyze requirements and design an electronic hardware/software solution using appropriate engineering tools.
2	Develop, integrate and test the designed system while addressing performance, functional and safety constraints.
3	Apply professional engineering pratices, documentation standards and project management techniques appropriate to industrial R&D environments.
4	Prepare technical documentation and deliver presentations effectively to diverse audience using modern communication tools.
5	Demonstrate creativity, teamwork and innovative thinking that contribute to entrepreneurial and societal needs.

PO -WK Mapping:

PO#	PO Description	Mapped WKs	PO#		Mapped WKs
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Environment and sustainability	WK5, WK7
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Ethics	WK7, WK9
PO3	Design / Development of Solutions	WK1, WK2, WK3, WK4, WK5, WK6, WK7	PO9	Individual & Team Work	WK9
PO4	Conduct Investigations of complex problems	WK2, WK3, WK4, WK8	PO10	Communication	WK7, WK9
PO5	Engineering Tool usage	WK2, WK6	PO11	Project Management & Finance	WK5, WK6, WK7
PO6	The Engineer and society	WK1, WK5, WK7	PO12	Life-long Learning	WK8

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	✓	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	✓	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES	✓	16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES	✓	17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

QUALITY ANALYSIS

Project collaboration (Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other
							✓
Project Guide, Department of Electronics and Communication Engineering							

Project Achievements:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes

S. No	Description	Details
1.	Prototype development	Yes
2.	Technology based solution developed	Yes
3.	Publication	-
4.	Start-up incubation	No
5.	Competition prizes	No
6.	Community / Industry / HEI service	Community
7.	SDG Indicators	3,9,10,11
8.	Grants / Financial Aid	No
9.	Multi-disciplinary	Yes
10.	Internship / Employment	No
11.	Self-employment skill	Yes
12.	Use of real-time data	Yes

Signature of the Guide **Signature of the Project Coordinator**